Current Electricity

Electric Current: (I) is defined as the net rate at which charge flows through a cross sectional area of a conductor. It is real a scalar quantity measured in Amperes (A).

$$I = dQ/dt$$

Current Density (J) The amount of current flowing per unit area of cross section of a conductor is called the current density.

$$J = I/A$$

For non-uniform surface, the current is the flux of the current density vector through the surface area:

$$I = \int J \cdot dA$$

** Electric Current and Current Density (J)

Current (I) is microscopic scalar quantity, but for engeneering (especially VLSL design), We use - current density (J), a vector feild.

High current density leads to electromigration, which can damage CPU traces.

Vector Form of Ohm's Law: $J = \sigma E$ (where σ is conductivity)

* Real-world Tip: If a wire is too thin (small Area) the current density (J) becomes very high, which makes the wire hot.

Ohm's Law

If there is a potential difference between the two ends of a conductor, electric current flows through it. The magnitude of this current depends on potential difference between the end of the conductor, shape and material of the conductor and the temperature of the conductor. If the temperature of a particular conductor and remains constant, then the current depends only on the potential difference of the two ends. George Simon Ohm (1786-1854) enuciated a law in this regard which is known as Ohm's law.

Statement of the Law: At a constant temperature current flowing through a particular conductor is directly proportional to the potential difference between the ends of the conductor.

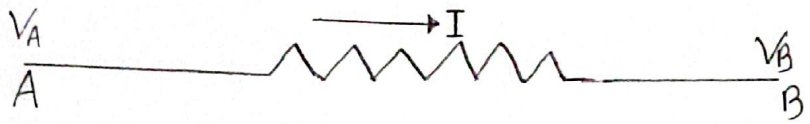


Fig: 1.1

Explanation: Let AB is a conductor, the potential at two ends of it are V_A and V_B respectively (Fig: 1.1). If $V_A > V_B$ then the potential difference of two ends would be $V = V_A - V_B$ and current will flow from A to B. Now, if the current flowing through the conductor at constant temperature is I, then according to Ohm's law.

$$I \propto V$$

Or, $I = GV$

Here, G is a proportionality constant known as conductance of the conductor. If we put the reciprocal of G. i.e, $R = \frac{I}{G}$ in the above equation we get,

$$I = \frac{V}{R}$$

Hence, $V = IR$

Where,

V = Potential difference between the two points of the conductor.

I = Current between the two points.

R = Electric resistance of the conductor

Explanation:

Drift Speed (V_d)

The average velocity with which the free electron move that is called drift velocity, V is known as Drift Speed (V_d).

$$J = nqV_d$$

Where:

n = Number of charge carries per unit volume (carrier density)

q = Charge of each carrier (approx. $1.6 \times 10^{19} C$ for e)

V_d = Drift velocity,

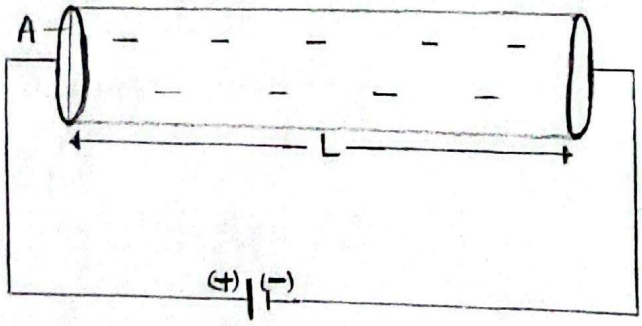
Hence, Drift Velocity,

$$V_d = J/nq$$

In a conductor, electrons move at high thermal speed ($\sim 10^8$ m/s) in random directions. When an electric feild is applied, they acquire a small net Drift speed ($\sim 10^{-4}$ m/s) opposite to the feild.

CSE CONNECTION: SIGNAL VS CARRIER SPEED The 'signal' (Information) in a copper trace travels at roughly 0.6 to 0.9 time speed of light because it is an electromagnetic wave. However, the physical electrons move slower tha a snail. This is why a computer responds instantly even though the electron are sluggish.

Derivation of $J = nqv_d$ or, $I = nAv_d$ or, $v_d = I/nA$



Let us consider a cylindrical conductor of cross-sectional area A and length L Let "n" be the number of free electrons per unit volume. That means 'n' is the number of density of electron inside the conductor. Hence

The volume of conductor, $V = A \times L$

The total number of electrons in the conductor,

$$N = n \times V = n \times (A \times L).$$

Total charge Q = (Total Electron) $\times$ (charge of one electron q)

$$= nALq.$$

If v_d is the drift speed, the time taken for all these charges to cross the surface of the conductor is

$$t = L / v_d.$$

Current(I) is the rate of flow of charge: $I = Q/t$ ..... (1)

Substituting Q and t in equation (1)

$$I = \frac{nALq}{t}$$

$$= nAq v_d \text{ ..... (2)}$$

Hence, $v_d = \frac{I}{nAq}$ ..... (3)

Since current Density

$$J = I/A$$

we get. $J = nAqv_d/A$

$$J = nAnqv_d$$

Hence, $v_d = \frac{J}{nq}$

Electromotive Force (EMF) and Power:

EMF ($\mathcal{E}$) is the work done per unit charge. It represent the energy per unit charge provided by a source (like a battery) to move charges through a circuit. It is measured in volts (V). Real batteries have Internal resistance (r).

Because real power source have internal resistance (r), the voltage measured across the terminals (V) is usally less than the EMF when a current (I) is flowing.

Terminal Voltage: $V = \mathcal{E} - Ir$

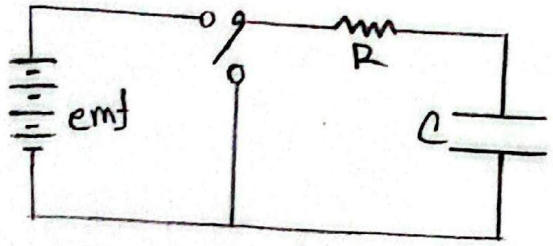
In mobile system, "Voltage sag" occurs when high CPU load draws current against this internal resistance.

Note: If the battery is being charged, the terminal voltage is $V = \mathcal{E} + Ir$.

* CSE CONNECTION: POWER MANAGEMENT Mobile and embedded system must account for r. As a battery drains, its internal resistance increases, causing a voltage sag, When high current is drawn by the CPU which can load to system instabillity for crashes

RC Circuits and Digital Timing

An RC circuit consist of a resistor (R) and a capacitor (C). These circuits are characterized by their time-dependent behavior.



Charging Capacitor

As the capacitor charges, the charge (q) increases exponentially toward it maximum value

$$q_t = C\mathcal{E}(1 - e^{-t/\tau})$$

Discharging Capacitor

When discharging through a resistor, the charge decays over time:

$$q_t = q_0 * e^{-t/\tau}$$

The Time Constant (τ)

The rate of charging of discharging is determined by the time constant:

$$\tau = R * C$$

- At $t = \tau$ a charging capacitor ~ 63.2% of its full charge
- At $t = \tau$ a dischargig capacitor retains ~ 36.8% of its initial charge.

** The RC circuit is the fundamental model for propagation deley in digital circuits. Every logic gate has an internal resistance and an output capacitance. The time it takes for a gate to switch depends on the time constant (τ).

Time Constant: $\tau = RC$

Charging Equation: $V_t = V_0(1 - e^{-t/RC})$

* Engineering Rule: The Rule of 5. After 5 times the time constant ($5 \times \tau$). We consider the capacitor 100% full. This "waiting time" is why computer have a maximum speed (clock speed). If we want a faster computer, we need smaller R and C.

- τ (Tau): The time required for the voltage to reach ~ 63.2% of its maximum.
- Switching Speed: In CSE, we typically wait 5τ (99.3% charge) to consider a digital signal as a stable "High".
- The 5GHz Limit: To increase CPU clock speed (frequency), we must decrease R and C to reduce the time constant, allowing the signal to switch faster.

CamScanner